CONCEPTUAL QUESTIONS

For Questions 1 through 3, interpret the position graph given in each figure by writing a very short "story" of what is happening. Be creative! Have characters and situations! Simply saying that "a car moves 100 meters to the right" doesn't qualify as a story. Your stories should make *specific reference* to information you obtain from the graph, such as distance moved or time elapsed.

1.

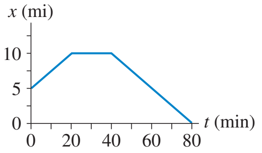


FIGURE Q2.1

2.

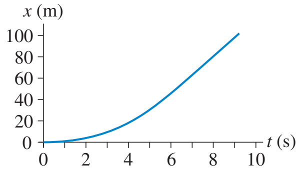


FIGURE Q2.2

3.

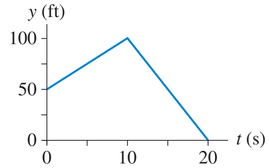


FIGURE Q2.3

4. FIGURE Q2.4 shows a position-versus-time graph for the motion of objects A and B as they move along the same axis.
   a. At the instant $t = 1$ s, is the speed of A greater than, less than, or equal to the speed of B? Explain.
   b. Do objects A and B ever have the *same* speed? If so, at what time or times? Explain.

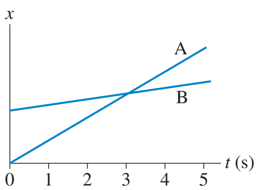


FIGURE Q2.4

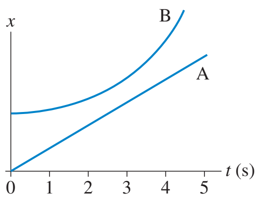


FIGURE Q2.5

5. FIGURE Q2.5 shows a position-versus-time graph for the motion of objects A and B as they move along the same axis.
   a. At the instant $t = 1$ s, is the speed of A greater than, less than, or equal to the speed of B? Explain.
   b. Do objects A and B ever have the *same* speed? If so, at what time or times? Explain.
6. FIGURE Q2.6 shows the position-versus-time graph for a moving object. At which numbered point or points:
   a. Is the object *moving* the slowest?
   b. Is the object moving the fastest?
   c. Is the object at rest?
   d. Is the object moving to the left?

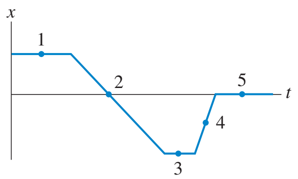


FIGURE Q2.6

7. FIGURE Q2.7 shows the position-versus-time graph for a moving object. At which numbered point or points:
   a. Is the object moving the fastest?
   b. Is the object moving to the left?
   c. Is the object speeding up?
   d. Is the object turning around?

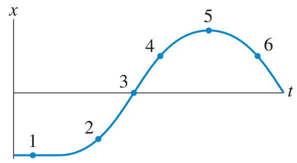


FIGURE Q2.7

8. FIGURE Q2.8 shows six frames from the motion diagrams of two moving cars, A and B.
   a. Do the two cars ever have the same position at one instant of time? If so, in which frame number (or numbers)?
   b. Do the two cars ever have the same velocity at one instant of time? If so, between which two frames?

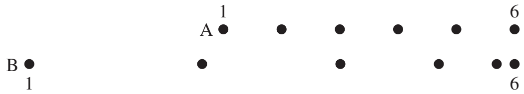


FIGURE Q2.8

9. You want to pass on a note to your friend who is traveling by a bus that does not stop in front of your house. You start jogging toward the bus the moment you see it at a distance. As the bus crosses you, do you think you can pass the note to your friend's outstretched hand?
10. When a space shuttle lands on a runway, it immediately deploys parachutes to reduce its tremendous speed. At this point, do the velocity and acceleration of the shuttle have the same direction? Explain.
11. Give an example of a motion
    a. where there is a positive acceleration, yet zero velocity.
    b. with zero acceleration but positive velocity.
12. You travel by car at a constant 90 km/h for 90 km. Then, due to heavy traffic, you need to reduce your speed to 50 km/h for another 100 km. What is your car's average speed for the 190-km trip?
13. A rock is *thrown* (not dropped) straight down from a bridge into the river below. At each of the following instants, is the magnitude of the rock's acceleration greater than g, equal to g, less than g, or 0? Explain.
    a. Immediately after being released.
    b. Just before hitting the water.
14. FIGURE Q2.14 shows the velocity-versus-time graph for a moving object. At which numbered point or points:
    a. Is the object speeding up?
    b. Is the object slowing down?
    c. Is the object moving to the left?
    d. Is the object moving to the right?

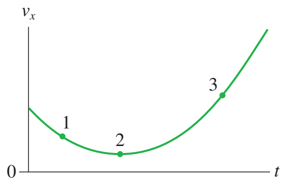


FIGURE Q2.14

EXERCISES AND PROBLEMS

Exercises

Section 2.1 Uniform Motion

1. || Larry leaves home at 9:05 and runs at constant speed to the lamppost seen in FIGURE EX2.1. He reaches the lamppost at 9:07, immediately turns, and runs to the tree. Larry arrives at the tree at 9:10.
   a. What is Larry's average velocity, in m/min, during each of these two intervals?
   b. What is Larry's average velocity for the entire run?

FIGURE EX2.1

2. || Julie drives 120 miles to her grandmother's house. She covers half the distance at 40 mph and the other half at 60 mph. On her return trip, she drives half the time at 40 mph and the rest at 60 mph.
   a. What is Julie's average speed on the way to her grandmother's house?
   b. What is her average speed on the return trip?
3. || Alan leaves London at 8:00 A.M. to drive to Leeds, 200 mi away. He travels at a steady 50 mph. Beth leaves London at 8:45 A.M. and drives a steady 60 mph.
   a. Who gets to Leeds first?
   b. How long does the first to arrive have to wait for the second?
4. || FIGURE EX2.4 is the position-versus-time graph of a bicycle. What is the bicycle's velocity at (a) $t = 5$ s, (b) $t = 15$ s, and (c) $t = 30$ s?

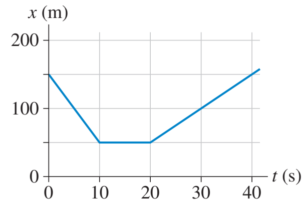


FIGURE EX2.4

Section 2.2 Instantaneous Velocity

Section 2.3 Finding Position from Velocity

5. | FIGURE EX2.5 shows the position graph of a particle.
   a. Draw the particle's velocity graph for the interval $0\text{ s} \le t \le 4\text{ s}$.
   b. Does this particle have a turning point or points? If so, at what time or times?

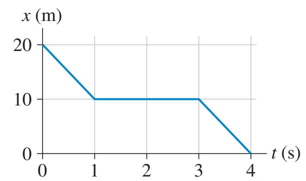


FIGURE EX2.5

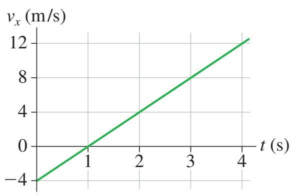


FIGURE EX2.6

6. || A particle starts from $x_0 = 10$ m at $t_0 = 0$ s and moves with the velocity graph shown in FIGURE EX2.6.
   a. Does this particle have a turning point? If so, at what time?
   b. What is the object's position at $t = 2$ s and 4 s?

7. | FIGURE EX2.7 shows the velocity graph for a particle having initial position $x_0 = 0$ m at $t_0 = 0$ s. At what time or times is the particle found at $x = 35$ m?

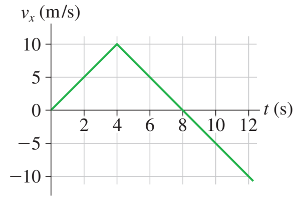


FIGURE EX2.7

8. || BIO FIGURE EX2.8 is a somewhat idealized graph of the velocity of blood in the ascending aorta during one beat of the heart. Approximately how far, in cm, does the blood move during one beat?

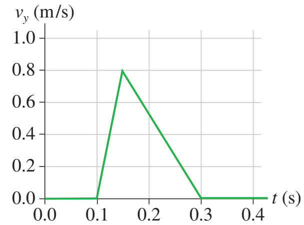


FIGURE EX2.8

Section 2.4 Motion with Constant Acceleration

9. | FIGURE EX2.9 shows the velocity graph of a particle. Draw the particle's acceleration graph for the interval $0\text{ s} \le t \le 4\text{ s}$.

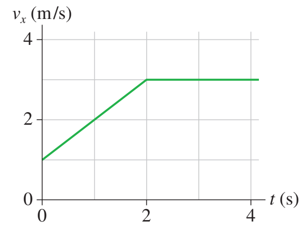


FIGURE EX2.9

10. || FIGURE EX2.10 shows the velocity graph of a particle moving along the x-axis. Its initial position is $x_0 = 2.0$ m at $t_0 = 0$ s. At $t = 2.0$ s, what are the particle's (a) position, (b) velocity, and (c) acceleration?

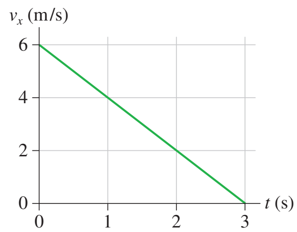


FIGURE EX2.10

11. || BIO FIGURE EX2.8 showed the velocity graph of blood in the aorta. What is the blood's acceleration during each phase of the motion, speeding up and slowing down?

12. FIGURE EX2.12 shows the velocity-versus-time graph for a particle moving along the x-axis. Its initial position is $x_0 = 2.0$ m at $t_0 = 0$ s.
 a. What are the particle's position, velocity, and acceleration at $t = 1.0$ s?
 b. What are the particle's position, velocity, and acceleration at $t = 3.0$ s?

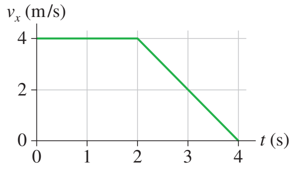


FIGURE EX2.12

13. a. What constant acceleration, in SI units, must a car have to go from zero to 60 mph in 4.9 s?
 b. How far has the car traveled when it reaches 60 mph? Give your answer both in SI units and in feet.
14. A jet plane is cruising at 280 m/s when suddenly the pilot turns the engines to full throttle. After traveling 4.0 km, the jet moves with a speed of 380 m/s. What is the jet's acceleration, assuming it to be a constant acceleration?
15. It has been proposed that a very small probe could be sent to a nearby star system by using a powerful laser beam, fired from an earth-orbiting satellite, to push on a lightweight "solar sail." Very high speeds could be reached in the vacuum of space by a fairly modest acceleration that continues for a long interval of time.
 a. Write an expression for the constant acceleration a_x an object needs to reach velocity v_{max} in time t_{push}, starting from rest.
 b. Write an expression in terms of v_{max} and t_{push} for the distance d the object travels during this time.
 c. For the mission to be feasible, the probe needs to reach 10% of the speed of light after being pushed for 1.0 year. The probe would then coast the rest of the way. What constant acceleration is needed? Note that the speed of light and much other useful data needed to solve problems are given inside the front and back covers of the book.
 d. What fraction of a light year will the probe have traveled at the end of the year? A light year (ly) is the distance traveled by light in 1 year.
16. BIO When you sneeze, the air in your lungs accelerates from rest to 150 km/h in approximately 0.50 s. What is the magnitude of the acceleration of the air in m/s^2?
17. A speed skater moving to the left across frictionless ice at 7.0 m/s hits a 5.0-m-wide patch of rough ice. She slows steadily, then continues on at 4.0 m/s. What is her acceleration on the rough ice?
18. A Porsche challenges a Honda to a 400 m race. Because the Porsche's acceleration of 3.5 m/s^2 is larger than the Honda's 3.0 m/s^2, the Honda gets a 1.0 s head start. Who wins? By how many seconds?
19. A Lamborghini Aventador S can go from 0 to 60 mph in 2.7 s. Assume the acceleration is constant.
 a. What is the magnitude of the acceleration?
 b. How far has the car traveled when it reaches 60 mph?

Section 2.5 Free Fall

20. Ball bearings are made by letting spherical drops of molten metal fall inside a tall tower—called a "shot tower"—and solidify as they fall.
 a. If a bearing needs 5 seconds to solidify enough for impact, how high must the tower be?
 b. What is the bearing's impact velocity?
21. A rock is tossed straight up from ground level with a speed of 20 m/s. When it returns, it falls into a hole 10 m deep.
 a. What is the rock's speed as it hits the bottom of the hole?
 b. How long is the rock in the air, from the instant it is released until it hits the bottom of the hole?
22. A student standing on the ground throws a ball straight up. The ball leaves the student's hand with a speed of 15 m/s when the hand is 2.0 m above the ground. How long is the ball in the air before it hits the ground? (The student moves her hand out of the way.)
23. A science project involves dropping a watermelon from the Empire State Building to the sidewalk below, from a height of 350 m. It so happens that Superman is flying by at the instant the watermelon is dropped. He is headed straight down at a speed of 40 m/s. How fast is the watermelon falling when it passes Superman?
24. BIO When jumping, a flea accelerates at an astounding 1000 m/s^2, but over only the very short distance of 0.50 mm. If a flea jumps straight up, and if air resistance is neglected (a rather poor approximation in this situation), how high does the flea go?
25. A rock is dropped from the top of a tall building. The rock's displacement in the last second before it hits the ground is 45% of the entire distance it falls. How tall is the building?

Section 2.6 Motion on an Inclined Plane

26. A car traveling at 30 m/s runs out of gas while traveling up a 10° slope. How far up the hill will it coast before starting to roll back down?
27. A skier is gliding along at 3.0 m/s on horizontal, frictionless snow. He suddenly starts down a 10° incline. His speed at the bottom is 15 m/s.
 a. What is the length of the incline?
 b. How long does it take him to reach the bottom?
28. Santa loses his footing and slides down a frictionless, snowy roof that is tilted at an angle of 30°. If Santa slides 10 m before reaching the edge, what is his speed as he leaves the roof?
29. A bicycle coasting at 7.0 m/s comes to a 6.0-m-long, 1.0-m-high ramp. What is the bicycle's speed as it leaves the top of the ramp?
30. A snowboarder glides down a 50-m-long, 15° hill. She then glides horizontally for 10 m before reaching a 25° upward slope. Assume the snow is frictionless.
 a. What is her speed at the bottom of the hill?
 b. How far can she travel up the 25° slope?

Section 2.7 Instantaneous Acceleration

31. FIGURE EX2.31 shows the acceleration-versus-time graph of a particle moving along the x-axis. Its initial velocity is $v_{0x} = 8.0$ m/s at $t_0 = 0$ s. What is the particle's velocity at $t = 4.0$ s?

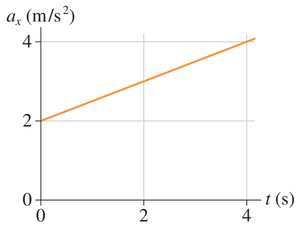


FIGURE EX2.31

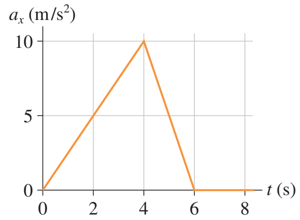


FIGURE EX2.32

32. FIGURE EX2.32 shows the acceleration graph for a particle that starts from rest at $t = 0$ s. What is the particle's velocity at $t = 6$ s?

33. | A particle moving along the x-axis has its position described by the function $x = (3.00t^3 - 3.00t + 5.00)$ m, where t is time (in seconds). At $t = 2.00$, what is
CALC
   a. the position of the particle?
   b. its velocity?
   c. its acceleration?

34. | A particle moving along the x-axis has its velocity described by the function $v_x = 2t^2$ m/s, where t is in s. Its initial position is $x_0 = 1$ m at $t_0 = 0$ s. At $t = 1$ s what are the particle's (a) position, (b) velocity, and (c) acceleration?
CALC

35. || The vertical position of a particle is given by the function $y = (t^2 - 4t + 2)$ m, where t is in s.
CALC
   a. At what time does the particle have a turning point in its motion?
   b. What is the particle's position at that time?

36. || The position of a particle is given by the function $x = (2t^3 - 6t^2 + 12)$ m, where t is in s.
CALC
   a. At what time does the particle reach its minimum velocity? What is $(v_x)_{min}$?
   b. At what time is the acceleration zero?

Problems

37. || Particles A, B, and C move along the x-axis. Particle C has an initial velocity of 10 m/s. In FIGURE P2.37, the graph for A is a position-versus-time graph; the graph for B is a velocity-versus-time graph; the graph for C is an acceleration-versus-time graph. Find each particle's velocity at $t = 7.0$ s.

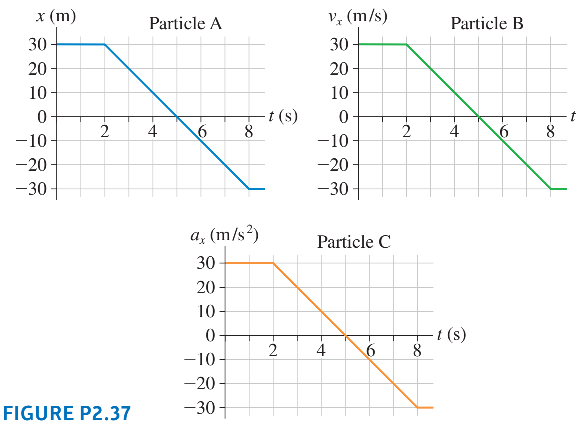


FIGURE P2.37

38. | A block is suspended from a spring, pulled down, and released. The block's position-versus-time graph is shown in FIGURE P2.38.
   a. At what times is the velocity zero? At what times is the velocity most positive? Most negative?
   b. Draw a reasonable velocity-versus-time graph.

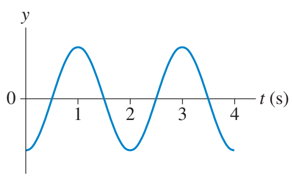


FIGURE P2.38

39. || A particle's velocity is described by the function $v_x = (t^2 - 10t + 21)$ m/s, where t is in s.
CALC
   a. At what times does the particle reach its turning points?
   b. What is the particle's acceleration at each of the turning points?

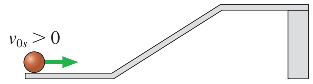
40. ||| A particle's velocity is described by the function $v_x = kt^2$ m/s, where k is a constant and t is in s. The particle's position at $t_0 = 0$ s is $x_0 = -9.0$ m. At $t_1 = 3.0$ s, the particle is at $x_1 = 9.0$ m. Determine the value of the constant k. Be sure to include the proper units.
CALC

41. || A particle's acceleration is described by the function $a_x = (10 - t)$ m/s^2, where t is in s. Its initial conditions are $x_0 = 0$ m and $v_{0x} = 0$ m/s at $t = 0$ s.
CALC
   a. At what time is the velocity again zero?
   b. What is the particle's position at that time?

42. || A particle's velocity is given by the function $v_x = (2.0 \text{ m/s}) \sin(\pi t)$, where t is in s.
CALC
   a. What is the first time after $t = 0$ s when the particle reaches a turning point?
   b. What is the particle's acceleration at that time?

43. || A ball rolls along the smooth track shown in FIGURE P2.43. Each segment of the track is straight, and the ball passes smoothly from one segment to the next without changing speed or leaving the track. Draw three vertically stacked graphs showing position, velocity, and acceleration versus time. Each graph should have the same time axis, and the proportions of the graph should be qualitatively correct. Assume that the ball has enough speed to reach the top.

FIGURE P2.43



44. || Draw position, velocity, and acceleration graphs for the ball shown in FIGURE P2.44. See Problem 43 for more information.

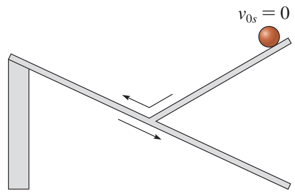
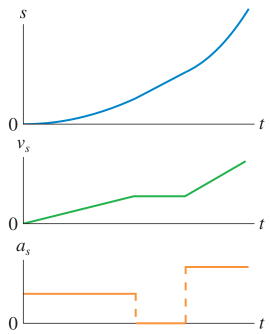


FIGURE P2.44

45. || FIGURE P2.45 shows a set of kinematic graphs for a ball rolling on a track. All segments of the track are straight lines, but some may be tilted. Draw a picture of the track and also indicate the ball's initial condition.



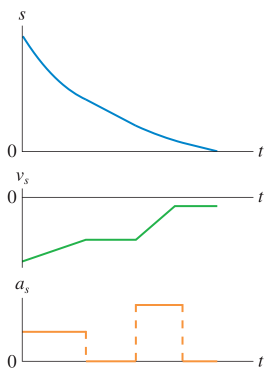


FIGURE P2.45 FIGURE P2.46

46. || FIGURE P2.46 shows a set of kinematic graphs for a ball rolling on a track. All segments of the track are straight lines, but some may be tilted. Draw a picture of the track and also indicate the ball's initial condition.

47. || You are driving to the grocery store at 20 m/s. You are 110 m from an intersection when the traffic light turns red. Assume that your reaction time is 0.50 s and that your car brakes with constant acceleration. What magnitude braking acceleration will bring you to a stop exactly at the intersection?

48. | The takeoff speed for an Airbus A320 jetliner is 80 m/s. Velocity data measured during takeoff are as shown.

t (s)	v_x (m/s)
0	0
10	23
20	46
30	69

a. Is the jetliner's acceleration constant during takeoff? Explain.
b. At what time do the wheels leave the ground?
c. For safety reasons, in case of an aborted takeoff, the runway must be three times the takeoff distance. Can an A320 take off safely on a 2.5-mi-long runway?

49. || You're driving down the highway late one night at 20 m/s when a deer steps onto the road 35 m in front of you. Your reaction time before stepping on the brakes is 0.50 s, and the maximum deceleration of your car is 10 m/s^2.
a. How much distance is between you and the deer when you come to a stop?
b. What is the maximum speed you could have and still not hit the deer?

50. || The Smooth Company has proposed transporting people between Paris and Amsterdam, a distance of 430 km, through an underground tube from which the air has been removed to eliminate air drag. Small pods carrying four passengers would accelerate at 2.5 m/s^2 until reaching a speed of 180 m/s. Later, they would brake at 1.5 m/s^2. A launch of one pod per minute would transport 240 passengers per hour, roughly equivalent to one jet plane per hour.
a. What would be the trip time in minutes from Paris to Amsterdam?
b. How far apart would two adjacent pods be on the constant-speed segment of the journey?

51. || A car starts from rest at a stop sign. It accelerates at 4.0 m/s^2 for 6.0 s, coasts for 2.0 s, and then slows down at a rate of 3.0 m/s^2 for the next stop sign. How far apart are the stop signs?

52. || A cheetah spots a Thomson's gazelle, its preferred prey, and
BIO leaps into action, quickly accelerating to its top speed of 30 m/s, the highest of any land animal. However, a cheetah can maintain this extreme speed for only 15 s before having to let up. The cheetah is 170 m from the gazelle as it reaches top speed, and the gazelle sees the cheetah at just this instant. With negligible reaction time, the gazelle heads directly away from the cheetah, accelerating at 4.6 m/s^2 for 5.0 s, then running at constant speed. Does the gazelle escape? If so, by what distance is the gazelle in front when the cheetah gives up?

53. || a. Find an expression for the minimum stopping distance d_{stop} of a car traveling at speed v_0 if the driver's reaction time is T_{react} and the magnitude of the acceleration during maximum braking is a constant a_{brake}.
b. A car traveling at 30 m/s can stop in a distance of 60 m, including the distance traveled during the driver's reaction time of 0.50 s. What is the minimum stopping distance for the same car traveling at 40 m/s?

54. || Bob is driving the getaway car after the big bank robbery. He's going 50 m/s when his headlights suddenly reveal a nail strip that the cops have placed across the road 150 m in front of him. If Bob can stop in time, he can throw the car into reverse and escape. But if he crosses the nail strip, all his tires will go flat and he will be caught. Bob's reaction time before he can hit the brakes is 0.60 s, and his car's maximum deceleration is 10 m/s^2. Does Bob stop before or after the nail strip? By what distance?

55. ||| A 1000 kg weather rocket is launched straight up. The rocket motor provides a constant acceleration for 16 s, then the motor stops. The rocket altitude 20 s after launch is 5100 m. You can ignore any effects of air resistance. What was the rocket's acceleration during the first 16 s?

56. || A 200 kg weather rocket is loaded with 100 kg of fuel and fired straight up. It accelerates upward at 30 m/s^2 for 30 s, then runs out of fuel. Ignore any air resistance effects.
a. What is the rocket's maximum altitude?
b. How long is the rocket in the air before hitting the ground?

57. || A lead ball is dropped into a lake from a diving board 5.0 m above the water. After entering the water, it sinks to the bottom with a constant velocity equal to the velocity with which it hit the water. The ball reaches the bottom 3.0 s after it is released. How deep is the lake?

58. || A hotel elevator ascends 200 m with a maximum speed of 5.0 m/s. Its acceleration and deceleration both have a magnitude of 1.0 m/s^2.
a. How far does the elevator move while accelerating to full speed from rest?
b. How long does it take to make the complete trip from bottom to top?

59. || Your car's anti-lock brake system is designed to keep the wheels from "locking" and starting to skid. The deceleration of a skidding car is less than that of a car that has the maximum braking without skidding—a topic we'll explore in Chapter 6. In one test, a car equipped with anti-lock brakes was able to decelerate at 7.0 m/s^2. while the same car without anti-lock brakes decelerated at 4.8 m/s^2 while skidding. In an emergency stop at a highway speed of 30 m/s, how much additional stopping distance would be needed by the skidding car compared to the car with anti-lock brakes?

60. || You are 9.0 m from the door of your bus, behind the bus, when it pulls away with an acceleration of 1.0 m/s^2. You instantly start running toward the still-open door at 4.5 m/s.
a. How long does it take for you to reach the open door and jump in?
b. What is the maximum time you can wait before starting to run and still catch the bus?

61. || Ann and Carol are driving their cars along the same straight road. Carol is located at $x = 2.4$ mi at $t = 0$ h and drives at a steady 36 mph. Ann, who is traveling in the same direction, is located at $x = 0.0$ mi at $t = 0.50$ h and drives at a steady 50 mph.
a. At what time does Ann overtake Carol?
b. What is their position at this instant?
c. Draw a position-versus-time graph showing the motion of both Ann and Carol.

62. || A steel ball rolls across a 30-cm-wide felt pad, starting from one edge. The ball's speed has dropped to half after traveling 20 cm. Will the ball stop on the felt pad or roll off?

63. || A very slippery block of ice slides down a smooth ramp tilted at angle θ. The ice is released from rest at vertical height h above the bottom of the ramp. Find an expression for the speed of the ice at the bottom.

64. || **FIGURE P2.64** shows a fixed vertical disk of radius R. A thin, frictionless rod is attached to the bottom point of the disk and to a point on the edge, making angle ϕ (Greek phi) with the vertical. Find an expression for the time it takes a bead to slide from the top end of the rod to the bottom.

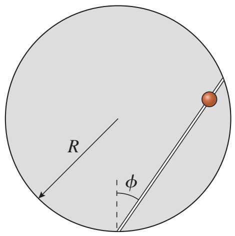


FIGURE P2.64

65. || A skateboarder starts up a 3.0-m-long ramp at 4.0 m/s. What is the maximum height of the ramp for which the skateboarder goes off the end rather than rolling back down?

66. || A motorist is driving at 20 m/s when she sees that a traffic light 200 m ahead has just turned red. She knows that this light stays red for 15 s, and she wants to reach the light just as it turns green again. It takes her 1.0 s to step on the brakes and begin slowing. What is her speed as she reaches the light at the instant it turns green?

67. || Nicole throws a ball straight up. Chad watches the ball from a window 5.0 m above the point where Nicole released it. The ball passes Chad on the way up, and it has a speed of 10 m/s as it passes him on the way back down. How fast did Nicole throw the ball?

68. || David is driving a steady 30 m/s when he passes Tina, who is sitting in her car at rest. Tina begins to accelerate at a steady 2.0 m/s^2 at the instant when David passes.
   a. How far does Tina drive before passing David?
   b. What is her speed as she passes him?

69. ||| If a Tesla Model S P100D in "Ludicrous mode" is pushed to
CALC its limit, the first 3.0 s of acceleration can be modeled as

$$a_x = \begin{cases} (35\text{ m/s}^3)t & 0\text{ s} \le t \le 0.40\text{ s} \\ 14.6\text{ m/s}^2 - (1.5\text{ m/s}^3)t & 0.40\text{ s} \le t \le 3.0\text{ s} \end{cases}$$

   a. How long does it take to accelerate to 60 mph? Your answer, which seems impossibly short, is confirmed by track tests.
   b. What acceleration would be needed to achieve the same speed in the same time at constant acceleration? Give your answer as a multiple of g.

70. ||| I was driving along at 20 m/s, trying to change a CD and not watching where I was going. When I looked up, I found myself 45 m from a railroad crossing. And wouldn't you know it, a train moving at 30 m/s was only 60 m from the crossing. In a split second, I realized that the train was going to beat me to the crossing and that I didn't have enough distance to stop. My only hope was to accelerate enough to cross the tracks before the train arrived. If my reaction time before starting to accelerate was 0.50 s, what minimum acceleration did my car need for me to be here today writing these words?

71. || As an astronaut visiting Planet X, you're assigned to measure the free-fall acceleration. Getting out your meter stick and stop watch, you time the fall of a heavy ball from several heights. Your data are as follows:

Height (m)	Fall time (s)
0.0	0.00
1.0	0.54
2.0	0.72
3.0	0.91
4.0	1.01
5.0	1.17

Analyze these data to determine the free-fall acceleration on Planet X. Your analysis method should involve fitting a straight line to an appropriate graph, similar to the analysis in Example 2.14.

72. ||| A ball is launched straight up at speed v_0. The second half of the total distance to the highest point is traveled during the final 1.0 s. How long does it take the ball to reach its maximum height?

73. || When a 1984 Alfa Romeo Spider sports car accelerates at
CALC the maximum possible rate, its motion during the first 20 s is extremely well modeled by the simple equation

$$v_x^2 = \frac{2P}{m}t$$

where $P = 3.6 \times 10^4$ watts is the car's power output, $m = 1200$ kg is its mass, and v_x is in m/s. That is, the square of the car's velocity increases linearly with time.
   a. Find an algebraic expression in terms of P, m, and t for the car's acceleration at time t.
   b. What is the car's speed at $t = 2$ s and $t = 10$ s?
   c. Evaluate the acceleration at $t = 2$ s and $t = 10$ s.

74. || Masses A and B in **FIGURE P2.74** slide on frictionless wires. They are connected by a pivoting rigid rod of length L. Prove that $v_{Bx} = -v_{Ay}\tan\theta$.

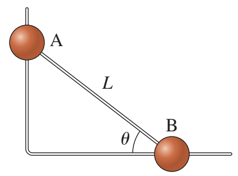


FIGURE P2.74

In Problems 75 through 78, you are given the kinematic equation or equations that are used to solve a problem. For each of these, you are to:
   a. Write a *realistic* problem for which this is the correct equation(s). Be sure that the answer your problem requests is consistent with the equation(s) given.
   b. Draw the pictorial representation for your problem.
   c. Finish the solution of the problem.

75. $64\text{ m} = 0\text{ m} + (32\text{ m/s})(4\text{ s} - 0\text{ s}) + \frac{1}{2}a_x(4\text{ s} - 0\text{ s})^2$
76. $(10\text{ m/s})^2 = v_{0y}^2 - 2(9.8\text{ m/s}^2)(10\text{ m} - 0\text{ m})$
77. $(0\text{ m/s})^2 = (5\text{ m/s})^2 - 2(9.8\text{ m/s}^2)(\sin 10°)(x_1 - 0\text{ m})$
78. $v_{1x} = 0\text{ m/s} + (20\text{ m/s}^2)(5\text{ s} - 0\text{ s})$
    $x_1 = 0\text{ m} + (0\text{ m/s})(5\text{ s} - 0\text{ s}) + \frac{1}{2}(20\text{ m/s}^2)(5\text{ s} - 0\text{ s})^2$
    $x_2 = x_1 + v_{1x}(10\text{ s} - 5\text{ s})$

Challenge Problems

79. ||| Water drops fall from the edge of a roof at a steady rate. A fifth drop starts to fall just as the first drop hits the ground. At this instant, the second and third drops are exactly at the bottom and top edges of a 1.00-m-tall window. How high is the edge of the roof?

80. ||| A rocket is launched straight up with constant acceleration. Four seconds after liftoff, a bolt falls off the side of the rocket. The bolt hits the ground 6.0 s later. What was the rocket's acceleration?

81. ||| A good model for the acceleration of a car trying to reach top
CALC speed in the least amount of time is $a_x = a_0 - kv_x$, where a_0 is the initial acceleration and k is a constant.
   a. Find an expression for k in terms of a_0 and the car's top speed v_{max}.
   b. Find an expression for the car's velocity as a function of time.
   c. A MINI Cooper S has an initial acceleration of 4.0 m/s^2 and a top speed of 60 m/s. At maximum acceleration, how long does it take the car to reach 95% of its top speed?

82. ||| Careful measurements have been made of Olympic sprinters
CALC in the 100 meter dash. A quite realistic model is that the sprinter's velocity is given by

$$v_x = a(1 - e^{-bt})$$

where t is in s, v_x is in m/s, and the constants a and b are characteristic of the sprinter. Sprinter Carl Lewis's run at the 1987 World Championships is modeled with $a = 11.81$ m/s and $b = 0.6887\ \text{s}^{-1}$.
   a. What was Lewis's acceleration at $t = 0$ s, 2.00 s, and 4.00 s?
   b. Find an expression for the distance traveled at time t.
   c. Your expression from part b is a transcendental equation, meaning that you can't solve it for t. However, it's not hard to use trial and error to find the time needed to travel a specific distance. To the nearest 0.01 s, find the time Lewis needed to sprint 100.0 m. His official time was 0.01 s more than your answer, showing that this model is very good, but not perfect.

83. ||| A sprinter can accelerate with constant acceleration for 4.0 s before reaching top speed. He can run the 100 meter dash in 10.0 s. What is his speed as he crosses the finish line?

84. ||| A rubber ball is shot straight up from the ground with speed v_0. Simultaneously, a second rubber ball at height h directly above the first ball is dropped from rest.
   a. At what height above the ground do the balls collide? Your answer will be an *algebraic expression* in terms of h, v_0, and g.
   b. What is the maximum value of h for which a collision occurs before the first ball falls back to the ground?
   c. For what value of h does the collision occur at the instant when the first ball is at its highest point?

85. ||| The Starship Enterprise returns from warp drive to ordinary space with a forward speed of 50 km/s. To the crew's great surprise, a Klingon ship is 100 km directly ahead, traveling in the same direction at a mere 20 km/s. Without evasive action, the Enterprise will overtake and collide with the Klingons in just slightly over 3.0 s. The Enterprise's computers react instantly to brake the ship. What magnitude acceleration does the Enterprise need to just barely avoid a collision with the Klingon ship? Assume the acceleration is constant.
Hint: Draw a position-versus-time graph showing the motions of both the Enterprise and the Klingon ship. Let $x_0 = 0$ km be the location of the Enterprise as it returns from warp drive. How do you show graphically the situation in which the collision is "barely avoided"? Once you decide what it looks like graphically, express that situation mathematically.